INSA Talent Center Department of Aerospace

Project proposal

Title : Geominer 1 Hybrid VTOL unmanned aerial vehicle for mineral exploration and mine site survey

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1. Executive summary

Exploration of mines and minerals often requires exploration of remote and hard to reach regions which makes the use of traditional ground based methods cumbersome, expensive and sometimes imposible. This project is aimed at development of a conceptual design and systems engineering of long endurance hybride VTOL unmanned aerial vehicle intended for mineral exploration, geological surveying, mine site mapping and other mining activities.

GEOMINER 1 is a hybrid VTOL fixed wing UAV system for autonomous and supervised exploration of minerals, geological surveys, mine site surveys, infrastructure surveys, and environmental surveys in remote and harsh terrain with limited infrastructure. It has a maximum take off weight (MTOW) of 30kg, a practical mission endurance target of 120 minutes, and a design mission range of 100km.

The airframe incorporates four VTOL lift motors, one rear cruise motor and one fixed wing of 2.8m wingspan. VTOL take off and landing are carried out using multirotor control upon reaching transition airspeed, the aircraft switches to fixed wing cruise mode, restoring endurance and range typical of traditional plane flight whilst still maintaining runway independence. The flight controller employs the ArduPilot QuadPlane firmware on a Pixhawk compliant standard open hardware platform mission planning, telemetry and post flight analysis use QGroundControl.

Navigation uses RTK GNSS combined with IMU, barometer, magnetometer and airspeed input by ArduPilot's EKF3 estimator. A modular payload bay with a quick releasable mechanical and electrical interface allows for interchangeable payload configurations with a total weight not exceeding 5kg survey grade LiDAR systems, RGB mapping cameras, hyperspectral and multispectral imagers, magnetometers, thermal cameras and multi gas sensors.

GEOMINER 1 is designed for operation in mountainous terrain, within bounded weather conditions,emergency water landing and takeoff for nearshore and off shore mineral exploration and for long distance autonomous survey missions, continuously supervised by the operator.

This document sets out the mission, requirements, architecture, physical design, propulsion, power supply, avionics, software, safety, manufacturing process and validation strategy to a degree adequate for an engineering team to undertake a detailed design, construct a prototype and conduct the test programme. Every design parameter that cannot be determined analytically is clearly labelled as a design target, an assumption or an experimentally validated value but never as a measured or known value.

Keywords: Hybrid VTOL UAV, Mineral Exploration, Geological Surveying, Mining, Pixhawk, QGroundControl, Long-Endurance UAV, Aerial Mapping.

2. Introduction

A number of activities in mining and mineral exploration involve collecting accurate information from extensive and hard to reach regions. Some of the tasks include geological surveys, mineral exploration, mine site mapping, terrain assessment, stockpile measurement, and environment monitoring. However, many mining areas are located in inaccessible, rugged, mountainous, and infrastructurally deprived areas where traditional surveying techniques are costly and may involve long periods of time and much human efforts. In addition, workers are at risk of encountering hazards resulting from unstable ground, dust and mining site weather conditions, among others.

Unmanned Aerial Vehicles (UAVs) offer an excellent tool for surveying such regions by air. UAVs have the capability of covering vast distances with reduced exposure to hazards and equipped with various sensors for mapping, imaging, geological survey, and environmental monitoring. However, traditional multi rotor UAVs usually suffer from limitations in terms of endurance and range since they are required to constantly create lifting forces from the rotors.

In turn, fixed-wing UAVs operate more efficiently during forward flights and have higher endurance and range as compared to multi rotor UAVs. Nevertheless, fixed wing UAVs usually need some kind of runway or launching and landing systems for their operation. However, a hybrid VTOL UAV could leverage the benefits of both the configurations. The aircraft could perform vertical takeoff and landing without the requirement of a conventional

runway, and also have an efficient fixed wing forward flight mode for long missions. Hybrid VTOL UAV configurations would therefore be ideal for mining and geological surveying operations that require a combination of flexibility and endurance in flight. Using a modular payload system, the aircraft would be able to perform different types of missions by using various sensors such as RGB cameras, multispectral and hyperspectral sensors, LiDAR, thermal cameras, and magnetometers.

Consequently, this project will be focused on developing and analyzing the conceptual design of a 30 kg long-endurance hybrid VTOL UAV for mineral exploration, geological surveying, mine site mapping, and other mining operations. The system will target an approximate endurance time of two hours, and 100 kilometers mission distance in remote areas, while still maintaining the VTOL capability. The design process will involve the consideration of airframe, hybrid propulsion system, power system, avionics, flight control, navigation, communication systems, and other aspects of the vehicle including payload systems and safety systems.

For this project, Pixhawk will be used as the flight-control platform and QGroundControl will be used for the ground control station. Engineering calculations, CAD modeling, aerodynamic and performance analysis, and necessary simulation techniques will be used to analyze the proposed configuration. The project will also address issues of component selection, system design, redundancy, safety, and reliability considerations. The goal of the project is therefore to design a technically feasible modular hybrid VTOL UAV that will help to achieve efficient and safer aerial data gathering for mining and geological surveying operations.

3. Background and Literature Review

UAVs have been increasingly adopted in mining and mineral exploration for purposes of geological mapping, mine site surveying, terrain modeling, stockpile measurements, and environmental monitoring. UAVs are able to acquire high resolution aerial data while at the same time minimizing time, cost and risks to personnel of surveying large and difficult terrains.

Multirotor UAVs exhibit very good VTOL performance as well as good hovering and low speed maneuverability. However, they suffer from lack of endurance due to the high power needed to

constantly hover. Fixed wing UAVs offer better aerodynamic performance and better endurance than multirotors in cruise flight mode although they need suitable launch and landing locations and cannot hover. This problem has triggered researchers to develop VTOL UAVs that combine the two modes of flight.

Existing hybrid VTOL systems are the quadplane, tiltrotor, tiltwing, and tailsitter configurations. All these systems show great promise in offering increased versatility while still retaining the endurance and range that are typical of fixed wing aircraft. Challenges however exist in areas such as control during transition, propulsion system design, energy optimization, payload, structural weight, and autonomous flying of the UAV.
In the application of UAVs to mining, the following additional requirements arise due to factors such as rugged terrain, weather conditions, dust, strong winds, lack of infrastructural facilities and the requirement for various types of survey sensors. For this reason, a modular UAV that carries interchangeable payload such as RGB camera, LiDAR, multispectral, thermal and magnetic sensor can increase mission flexibility.

With these issues in mind, this project seeks to conceptually design and perform systems engineering of a 30 kg hybrid VTOL UAV with an endurance of about 2 hours and mission range of about 100 km. The systems involved in this UAV design will include airframe, propulsion, power, avionics, navigation, communication, flight control and payload systems. Pixhawk and QGroundControl will be included in the flight control and ground control systems respectively. Engineering calculations, CAD and simulation will be used to assess the proposed design.

4. Problem statement

- ✓ Surveying and Mapping limitations

Exploration of mineral resources and mineral extraction operations require the accurate and efficient surveying and mapping of expansive regions for geological evaluation, mine planning, monitoring, and resource management. Traditional forms of ground based surveying, may prove to be inefficient in terms of time and effort especially when surveying expansive and inaccessible regions.

- ✓ Harsh and Hazardous mining Enviroments

The mining industry often operates in dangerous mining environment conditions where the topography is rough, there is dust, water, temperature change, and wind. Such conditions might be detrimental to surveying devices and compromise the reliability of operations. Moreover, exposing people to dangerous sites such as rocky slopes will be unsafe and decrease the efficiency of traditional surveys.

- ✓ Limitations of existing surveying technologies

Current UAV platforms offer certain capabilities that can be utilized in mining and geologic surveys however, these UAV platforms also have several limitations. Multirotor UAVs are capable of doing vertical takeoff and landing, yet tend to be of low endurance and range, while fixed wing UAVs offer higher flight efficiency, but also need to have certain areas for takeoff and landing. Standard UAVs, moreover, tend to be made to work on land only and are likely to be unable to land in case of an emergency landing in the water. Besides that, most standard aerial surveying platforms are not built for underwater research and data collection from water environment.

4.1 Problems That This System solves

GEOMINER 1 offers personnel independence, endurance, and modular multi mission sensing in a single airframe. This system enables people to be removed from dangerous environments such as toxic gases, unstable surfaces, and remote locations it condenses survey cycles from days to hours and creates datasets that can be used for resource evaluation, pit designing, stockpile quantifying, and environmental monitoring.

The VTOL design fills the gap between the multirotor agility and fixed wing efficiency. The aircraft is capable of launching and landing in vertical position from any flat surface including water surfaces and transition to wing supported cruising most of the time during the flight using much less energy compared to an aircraft of similar weight in a multirotor configuration.

4.2 What This System Does Not Solve

- ✓ Ore grade certification is not performed in real time as sensor readings still need interpretation and, when required, validation in the lab.

5.Aim and objective of project

5.1. Aim

The goal of this project is to develop and evaluate the design of a long endurance hybrid VTOL UAV weighing 30 kilograms, suitable for mineral discovery, geological survey, and mining applications, capable of taking off and landing vertically, efficient long-range flying with fixed wings, multi sensor data collection, and improved operational safety.

5.2. Objectives of the Project

1. Establish the system and mission requirements of 30 kg long endurance hybrid VTOL UAV designed for mineral exploration, geological surveying, mine site mapping and other mining operations.
2. Develop the conceptual airframe design of the hybrid VTOL UAV consisting of the wings, fuselage, VTOL system, propulsion system, landing system, payload bay, and structural configuration.
3. Design the aircraft systems and sub systems, including propulsion, power, flight control, navigation, communication, avionics, payload and structural systems.
4. Identify the suitable components and technologies in terms of the engineering requirements, performance, reliability, weight, power requirements and compatibility of the aircraft, including the use of the Pixhawk autopilot and QGroundControl ground control station.
5. Perform aerodynamic, propulsion, weight, stability and performance analysis to examine if the proposed UAV is able to meet approximately 2 hours endurance, 100 km range, and desired VTOL and cruise performance.
6. Establish the UAV system architecture and develop the block diagram showing the interaction between the major systems and sub-systems of the aircraft.
7. Develop and evaluate the design of the proposed UAV using the engineering software and simulation tools and identify the software utilized throughout the design process.
8. Develop a modular multi mission payload system enabling different sensors for mineral

exploration, geological surveying, mapping and environmental monitoring.

9. Perform safety and redundancy analysis and examine mission reliability for flying the UAV in rugged, remote and mountainous environment with limited infrastructure.

10. Perform the engineering trade off and performance analysis of the UAV design to determine suitable configuration and demonstrate the mission capabilities of the proposed hybrid VTOL UAV.

6.Research and scope

This research concerns the conceptual design and development of a VTOL fixed wing UAV for the exploration and mining of minerals. This research will analyze airframe configuration, aerodynamics, propulsion, power source, flight control system, navigation, communication, payload, and safety features. The research will also look at sensors like LiDAR, RGB, multispectral, hyperspectral, thermal, magnetometer, and gas sensors for geological surveys and environmental monitoring.

The scope of this project includes the design of a UAV that has vertical take off and landing ability, efficient cruising in a fixed wing, navigation, and long endurance. This project also covers designs that operate in challenging environments like dusty, windy conditions, temperature changes, high altitudes, and little ground infrastructure. The project also explores future advancements of this project including AI detection of minerals, swarm operation, satellite communication, solar energy generation, weatherproof design, redundant flight controllers, high capacity batteries, and ability to fly in water.

This research is scoped to the conceptual design and preliminary engineering analysis of the UAV. Manufacturing, full scale testing, certification, and future development of new technology like solid state battery, satellite system, and underwater ability are out of scope.

7.Mission Definition and Operational Concept

7.1 Mission Statement

Aerial survey of mineral exploration concessions, open pit and underground mine sites (external

surfaces and entrances), and related infrastructure, collecting high-resolution geospatial, spectrographic, magnetic, and environmental data to be interpreted by human geologists.

7.2 Typical Mission Profile

1. Site survey and geofence creation. The operator surveys the launch/recovery site, clears the flight path, and establishes a virtual geofence in QGroundControl software.

2. Aircraft and payload pre-flight inspection. Battery status is confirmed, the payload module is verified as correctly loaded and secured, control surface functionality tested, and propellers inspected.

3. RTK base station placement and communications test. The RTK GNSS base station is placed above the known site, the communications link is verified, and the integer ambiguity solved prior to launch.

4. Vertical lift off under operator control. The aircraft is lifted vertically from the ground to hover.

5. Transition to fixed-wing cruise. Under operator control or autonomous trigger, the aircraft accelerates to transition speed (target 15-17m/s); the VTOL motors become idle, the pusher takes over thrust for cruise.

6. Autonomous collection of survey grid/corridor. The aircraft flies autonomously according to pre-programmed survey pattern in cruise, collecting payload data continually while being monitored by the operator through QGroundControl.

7. On-station loiter or re-tasking. An area of interest may be ordered to loiter, or the aircraft may be ordered to execute a new pattern of waypoints.

8. Return transition to VTOL control. At the end of the survey leg, the aircraft decelerates, the pusher motors reduce thrust, and VTOL motors spool up to assume multirotor control.

9. Vertical landing at the designated recovery site. The aircraft lands vertically under operator control using RTK positioning.

10. Data download and integrity check. Data from the payload is downloaded to the ground compute node; a check is made of integrity of the dataset.

11. Human analysis and report generation. The data are transformed into orthomosaic images or point clouds or spectra for geological interpretation.

7.3.Supervision Scheme

The take off, landing, transition maneuvers, as well as any emergency action is subject to supervision. Monitoring and emergency stop capabilities rest with the operator during the flight process. Survey path generation occurs independently with ongoing supervision by the operator through the QGroundControl software. Override, hold, return to launch (RTL), and termination capabilities remain with the operator throughout the flight process.

8.Oprating Envelope

Parameter	Value	type	Basis
Flight condition	Daylight VMC only	requirement	No light vision or de-icing systems are included in the current desgn
Max**imum** density altitude	4000m ASL	assumption	Propulsion and wing performance limit.
Sustained wind	≤12m/s	Design target TBD	Requires flight test confirmation
Wind gusts	≤15m/s	Design target TBD	Momentary control authority margin.
Operating temperature	−5 to +40°C	Assumption	Based on preliminary Battery and avionics

			thermal limits.
Normal take off and landing surface	Land	Requirement	Normal VTOL takeoff and landing are performed on suitable land surfaces
Emergency water landing	Permitted	Requirement	The UAV shall be capable of landing on and remaining afloat on water surface
Terrain slope at landing or launch area	$\leq 15^{\circ}$	Assumption	Based on Preliminary Landing gear and VTOL stability point limit.

Table 1.1: Operating Envelope Parameters

9.System Requirements

- ✓ Every parameter listed has its own clear definition as a requirement (a set requirement fulfilled by the design)
- ✓ design target (the optimal value targeted by the design, which is bounded but not finalized)
- ✓ derivation (a calculated value based on the requirements)
- ✓ assumption (a given condition in this particular design)
- ✓ TO be defined (values that cannot be specified without physical testing).

This classification system is used in all the tables provided so that none of the assumptions or calculations can pass for measurements.

Parameter	Value	Type	basis
MTOW 30kg	30Kg	Requirement	Mass budget sized for 120 min edurance and 100 km range
Payload capacity	up to 5kg	Design target	Modular bay sized for survey LiDAR or hyperspectral packages
Practical mission endurance	≥120min	Requirement	Minimum acceptablefor largeconcession surveys.
Mission range	100km	Design target	Derived from cruise power and battery energy after reserves.
Cruise speed	18-25 m/s	Design target	Selected for mapping GSD and endurance balance.
Maximum speed	30 m/s	Design target	Structural and control margin limit.
Transition airspeed	15–17m/s	Design target	QuadPlane transition corridor: above stall, below max structural load limit.
Hoverthrust/weight ratio	≥1.8	Requirement	Control authority for VTOL in moderate wind and on sloped terrain.
Wingspan	2.8m	Design target	Aspect-ratio and structural trade off transportability limit.

Overall length	≈1.9m	Derived	Fuselage, payload bay, and tail volume packaging.
Height (gear to top)	≈0.55m	Design target	Propeller clearance and transportability.
RTK horizontal accuracy	<1cm	Requirement	Mapping grade georeferencing for stockpile and pit measurements.
GSD (mapping camera)	<1cm	Design target	At typical survey altitude of 80–120m AGL.
Service ceiling	4000m ASL	Assumption	Density-altitude limit for propulsion and aerodynamics.
Wind limit (sustained)	12m/s	Design target / TBD	Requires flight-test confirmation at full MTOW.
Wind gusts	15m/s	Design target / TBD	Availablecontrol authority at selected hover T/W.
Operating temperature	−5 to +40°C	Assumption	Typicalremoteexploration climatesbattery chemistry limit.
Battery chemistry	Li-ion (14S)	Requirement	Energy density For mass budget closure.
Charging	External charger only	Requirement	Battery removed or connected externally,no onboard charging.

Communication link range	≥20km LOS	Design target	Terrain dependent representative of large concession line of sight.
Telemetry latency	<500ms round trip	Design target	Standard MAVLink radio performance.

Table1.2: system requirements

10. System Architecture

GEOMINER 1 is made up of seven key systems:

1. Structural system:airframe, wing, tail booms, empennage, landing gear, VTOL arm assemblies.

2. Avionics and flight control system : Pixhawk flight controller, IMU, RTK GNSS, magnetometer, barometer, airspeed sensor, obstacle sensors.

3. Propulsion system : four VTOL lift motors, one cruise pusher motor, ESCs, propellers.

4. Electrical power system : main battery, backup battery, BMS, PDB, wiring, fuses.

5. Mission computing and communications system : companion computer, telemetry radio, data storage, long-range communication channels.

6. Payload system : modular payload bay, quick-release mount, three-axis gimbal, sensor modules.

7. Safety and emergency system : ballistic recovery parachute, flight termination system (FTS), emergency locator beacon, geofencing logic.

10.1 Structural system

Structure is the physical structure of the drone, which supports propulsion system, electronics, battery, payload, and other components, while also providing protection during flight, landing and water operations.

1. Central Fuselage

The central part of the drone. It houses such critical components as the battery, flight controller, avionics, wiring, and payload electronics. In addition, it provides the main connection between the wings, propulsion system, and other components.

2. Motor Arms

Arms that are connected to the main body and mount the VTOL motors and propellers. They transmit the power and vibrations from the motors to the main airframe.

3. Floats or Hulls

Devices for flotation attached to the bottom of the drone.These devices ensure buoyancy, preventing the drone from sinking in the water, and keep the main body of the drone free from water contact.

5.Landing Gear

Supports the drone upon landing and staying on the ground. They also keep the fuselage, payload, and other sensitive components away from the ground during landing.

6. Payload Bay

The bay that provides for mounting and securing the mining sensors and cameras, such as LiDAR, RGB, multispectral, and thermal cameras. The bay should be easily accessible to install, remove or change the payload.

7.Protective Closures

Protective closures for the sensitive parts of the drone to protect from dust, moisture, debris, and impact.These protective closures are particularly important since the drone will operate in the conditions of mining and water environments.

8. Seals and Gaskets

Sealing elements mounted around doors, access panels, connectors, and other openings. They provide better environmental protection by preventing water and dust intrusion into the electronics compartments.

9. Structural Material

Material chosen based on strength, weight, durability, and environmental resistance.

- ✓ Carbon fiber provides excellent strength to weight ratio
- ✓ Fiberglass provides good environmental resistance and durability
- ✓ Aluminum can be used as reinforcement
- ✓ Reinforced plastics and corrosion resistant fasteners provide additional resistance to water and dust in harsh mining environment.

10.2. Avionics and flight control system

The avionics and flight control system are the electronic devices which control the drone, determine its position and orientation, monitor its health state, and provide stable flight.

1. Flight Controller / Autopilot

The central computer for control of the drone. Processed sensor data and sends control commands to the motors and other control systems for stabilization and maintenance of the right altitude, orientation, and flight mode.

2. IMU (Inertial Measurement Unit)

It determines the movement and orientation of the drone. In most cases, it represents the data obtained by gyroscope and accelerometer sensors.

3. Gyroscope

It measures the drone's angular rotation around the three axes of rotation. Thus, it enables the flight controller to know if there is any change in roll, pitch, and yaw.

4. Accelerometer

It measures the drone's linear acceleration and movement along its three axes of motion. Provides information about changes in the drone's movement and attitude.

5. Barometer

It measures atmospheric pressure to find out the drone's altitude. Used mainly to control and maintain the drone's vertical position.

6. Magnetometer

It acts as the digital compass by determining the Earth's magnetic field. The information from magnetometer helps to determine the drone's heading and improve its navigation.

7. GNSS / GPS Receiver

Provides the position, ground speed, and navigation information for the drone. Used for navigation and waypoint control.

8. RTK Module

It is used to determine the position with high precision by correcting errors in GNSS. This module can be used for mineral exploration and mapping.

9. Radar / LiDAR Altimeter

It measures the distance from the drone to the ground or water surface. Needed for low altitude flying and maintaining a certain distance to the surface.

10. Optical Flow Camera

The camera for detection of relative movement between the drone and the surface. Provides position and stability at low altitude in conditions of weak GPS signals.

11. Motor Control Outputs

The electrical control signals from the flight controller to the ESC. These signals regulate the thrust level of each motor.

12. Water Contact Sensors

They indicate if the floats or hulls of the drone touch the water surface. Useful information to switch between flight and water operation.

13. Surface Navigation Sensors

They provide information about the drone's heading, movement, and position on water. Used for controlled movement on water surface.

10.2.1. Flight Control and Ground Control Software

The flight controller runs ArduPilot QuadPlane firmware on Pixhawkstandard open hardware.

Reasoning : Four particular characteristics of this VTOL fixed-wing platform underlie this decision:

1. QuadPlane support. ArduPilot includes a well tested production set of flight modes that considers vertical take off, hover, forward transition, cruise, back transition, and vertical landing as a unified, parameterized chain. The transition logic is well documented and validated via simulation and actual flights.

2.RTK precision sensor fusion. EKF3 estimator of ArduPilot fuses IMU, RTK GNSS, barometer, magnetometer and airspeed in one state estimate, which can support sub-5cm positioning requirements during survey passes and precision landing.

3. Open architecture for simulation. Fully functional software-in-the-loop (SITL) simulation ensures transition logic, fail-safe behaviour and mission sequencing validation before hardware implementation, thus eliminating the project risks from the very beginning.

4. Survey primitives for long endurance fixed wing missions. ArduPilot already contains all necessary mission items (survey grid, corridor, loiter, terrain following) that fit mineral exploration mission scenario without any additional waypoint generator development. Considered alternative. PX4 supports VTOL capability in a quite capable manner, yet the mode

transition state machine is a bit rigid and has less documented examples of long-endurance fixedwing survey at 30kg class. ArduPilot QuadPlane is our autopilot of choice.

Ground Control Station

Q GroundControl connects to the ArduPilot flight controller through MAVLink and serves three purposes in the GEOMINER 1 system:

- ✓ **Pre-flight:** defining the survey grid from satellite imagery, geofencing, RTL altitude, airspeed/sensor calibration parameters, and mission verification.
- ✓ **Flight:** real time telemetry display (position, orientation, battery, airspeed, RTK fix), real-time mission modification (waypoint retasking), manual intervention using joystick or pointing and clicking "Go To" missions, payload management (camera trigger/gimbal orientation), and audio alerts.
- ✓ **Post-flight:** logging extraction, EKF/power usage analysis, mission reconstruction and creation of KML/GeoTIFF files for GIS use.

QGroundControl is also the primary platform utilized in the ArduPilot SITL simulation testing during the design process.

10.3. Propulsion system

It produces the thrust needed to make the drone take off, fly, hover, maneuver, and work on the water surface. In your amphibious VTOL drone, the system includes aerial propulsion and water propulsion.

Aerial Propulsion

- ✓ Brushless Motors

  Convert the energy from the battery into rotation. Motors rotate the propellers to create thrust needed to take off vertically, hover, and fly.
- ✓ Propellers

  Convert rotation of motors into thrust by the air. They influence lift, efficiency, flight speed, and energy consumption of the drone.
- ✓ Electronic Speed Controllers (ESCs)

  Manage speed of each brushless motor based on commands from the flight controller.

Controlling speed of individual motors makes it possible to control the altitude, roll, pitch, and yaw.

- ✓ Motor Mounts

  Attach each motor to the motor arm or airframe. They have to be strong enough to withstand vibrations, thrust, and loads.

- ✓ ESC Telemetry

  Provides data on current of each motor, voltage, temperature, and rotation speed. It allows detecting abnormal working of motor or ESC.

Water Propulsion

- ✓ Water Thrusters

  Create thrust to make the drone move across the water surface. It helps it to move forward, backward, or change direction being on the water.

- ✓ Thruster Controllers

  Manage the speed and direction of water thrusters based on commands from the control system.

- ✓ Water Mode Controller

  Controls transition between aerial flight and water surface mode. It ensures that appropriate propulsion system and control functions are active in each mode.

- ✓ Surface Navigation Sensors

  Provide information about heading, navigation, and location on the water surface.

- ✓ Emergency Thruster Cutoff

  Turns off water thrusters in case of an obstruction, electrical faults, or other emergencies.

10.3.1. Thrust Requirement

At 30kg MTOW the aircraft weight is:

$$W = m \cdot g = (30\text{kg})(9.81\text{m/s}^2) = 294.3\text{N}. \quad (1)$$

The hover thrust to weight ratio is set to 1.8 to provide adequate control authority for VTOL in moderate wind and on sloped terrain:

$$T\text{hover,total} = W = 294.3\text{N} \quad (2)$$

$$T_{\max,\text{total}} = 1.8 \cdot W = 529.7\text{N} \quad (3)$$

$$T_{\text{hover,motor}} = \frac{294.3}{4} = 73.6\text{N (7.5kgf)} \quad (4)$$

$$T_{\max,\text{motor}} = \frac{529.7}{4} = 132.4\text{N (13.5kgf).} \quad (5)$$

10.3.2. Motor and Propeller Selection

A per motor hover thrust of 7.5kgf falls within the documented performance envelope of largeframe multirotor motors driving 22-inch propellers on 14S (51.8V nominal) power the same motor and propeller class used across heavy lift commercial multirotor and survey platforms.

Parameter	Selection
VTOL motor class	Large frame multirotor, ≈350g each, rated for 22in propeller at 14S
VTOL propeller Diameter	22in (0.559m)
VTOL propeller pitch	≈8in , final selection is a bench-test item
VTOL propeller type	Two blade, fixed pitch, folding optional for drag reduction
Cruise motor class	Fixed wing outrunner or inrunner, ≈400g, rated for 18in folding propeller
Cruise propeller diameter	18in (0.457m), folding two-blade
ESC rating (VTOL)	60A continuous minimum, 14S rated, per motor
ESC rating (cruise)	80A continuous, 14S rated
Battery voltage	14S Li-ion, 51.8V nominal, 58.8V fully charged, 42.0V lowvoltage cutoff

Table 1.3: motor and propeller selection

10.3.3. Hover and Cruise Power

10.3.3.1.Hover Power (Momentum Theory Estimate)

Hover power is estimated using actuator-disk (momentum) theory, corrected by a figure of merit (FM) accounting for real propeller losses. A conservative FM of 0.75 is used for the large, slow-turning VTOL propellers.

$$P_{\text{ideal}} = \frac{T^{3/2}}{\sqrt{2\rho A}}$$

where T = 73.6N per rotor, ρ = 1.225kg/m^3 (standard sea level), and $A = \pi(0.559/2)^2$ = 0.245m^2. (6)

Parameter	Value
Propeller radius	0.2795m
Disk area per rotor	0.245m^2
Ideal power per motor	813W
Figure of merit	0.75
Actual power per motor	1084W
Total VTOL hover power (4 motors)	4336W

Table 1.4: hover power estimate

10.3.3.2 Cruise Power Estimate

Cruise power is derived from the XFLR5 drag polar for the complete aircraft at the design cruise speed of 20m/s. XFLR5 analysis at the target Reynolds number gives a total drag coefficient $C_D \approx 0.095$ for the clean configuration.

$$D = \frac{1}{2}\rho V^2 S C_D \approx 65 \text{N} \quad (7)$$

where $S \approx 0.80$m^2 is the wing reference area. Required cruise power at the propeller, at a propulsive efficiency $\eta_p \approx 0.72$:

$$P\text{cruise} = \frac{D \cdot V}{\eta_p} = \frac{65 \times 20}{0.72} \approx 1806\,\text{W} \quad . \quad (8)$$

10.3.3.3. Total System Power

Total mission power includes propulsion, avionics, compute, and payload. The full breakdown is given in the Power Budget and Endurance section.

Item	Power
VTOL propulsion, 4 motors, hover	4336
Cruise propulsion, pusher motor	1806W (design baseline)
Avionics, compute, sensors, communication	155W
Payload (typical survey load)	85W

Table 1.5 : total system power

The 14S bus voltage keeps total current manageable: hover current ≈84A total (21A per VTOL motor); cruise current ≈38A at the main bus. This holds wiring, connector, and ESC current ratings within practical commercial ranges.

10.4.Electrical power system

The electrical power system generates, transfers, controls, and protects electricity across the UAV. It provides electricity to the motors, ESCs, flight controller, avionics, communication equipment, sensors, and mining payload.

1. Main Battery

The primary source of energy for the UAV. Provides electrical power to the propulsion system and other electronic systems. For the long-endurance UAV, the battery capacity should be chosen in accordance with the amount of energy required for flying.

2. Battery Management System (BMS)

It monitors and protects the battery by tracking the battery voltage, current, temperature, and condition. It prevents such issues as overcharging, over discharging, and overheating.

3. Power Distribution Board (PDB)

The board distributing electrical power from the main battery to other systems like ESCs, flight controller, avionics, communication equipment, and payloads.

4. Voltage Regulators

Regulating the voltage of the battery into proper voltages for different electroniccomponents. For example, some components, such as avionics, require lower and more stable voltage than the propulsion system

5.Current Sensor

Measuring the electric current drawn by the UAV. The information is used by the flight controller to determine the power consumption and the amount of remaining battery charge.

6. Voltage Sensor

Measuring the voltage of the battery and systems. It monitors the battery condition and provides the information about low voltage conditions for the flight controller.

7. Power Switches and Fuses

Provide the ability to turn on/off the electrical system and to protect it from short circuits, excessive currents, and other electrical faults.

8. Backup Power Supply

A system providing power to the critical components like the flight controller, navigation, and communication equipment in case of a power system malfunction. Thus, it provides the UAV with a possibility to make an emergency reaction, for instance, return to a safe place.

Power flow

The general power flow may be presented as:

Main Battery → PDB → ESCs → Motors

Main Battery → Voltage Regulators → Flight Controller / Avionics / Communication / Payloads

BMS, voltage sensor, and current sensor monitor the electrical system permanently; fuses, switches, and backup power supply provide protection and redundancy.

For our vertical takeoff and lnding mining UAV

- ✓ Battery

14S lithium ion pack is selected for energy density. Capacity requirement is calculated based on the mission energy budget (Section 23). Installed mass is estimated to be 10.0kg, or about 33percent of MTOW. The battery pack is installed in a removable tray located at the wing fuselage junction, fastened with quick-release latches for fast ground battery swap during sorties.

- ✓ Power Distribution Board

Custom or commercial PDB receives input from the primary battery pack and distributes the power supply to the five ESCs, measuring the current on the main bus for reporting the battery management information. Another output, regulated and isolated, supplies the avionics and payload secondary bus.

- ✓ Bus Isolation

  Avionics, companion computer and the sensor payload receive electrical power via an additional DC to DC converter that is supplied from the primary battery, but electrically isolated from the motor power bus. The electrical isolation of the buses prevents motor switching transients from being coupled into the payload's sensitive analog and digital circuitry.

- ✓ Electronic Speed Controllers

  Four separately addressed VTOL ESCs (60A continuous rating, 14S), and one cruise ESC (80A, 14S) are mounted in the configuration that provides sufficient space for convective cooling. The ESCs are mounted close to the motors to provide shortest possible power leads.

- ✓ Battery Management and Failsafe

  Individual cell voltages and pack temperature are measured by the BMS and provided to the flight controller. When a preset low voltage is reached, the flight controller initiates the low-battery RTL mode, taking over from the current mission command. Low voltage threshold is selected with safety margin above the 42.0V cutoff, enough to ensure safe return and landing from the max operating distance.

- ✓ Backup Power

  Small Li ion battery pack (about 0.2kg) keeps the flight controller, RC receiver and telemetry radio running in case of main battery pack failure or BMS disconnect. It does not supply power to propulsion system.

10.5.Mission computing and communications system

The mission computing and communication system controls data processing, mission planning, data storage, and communication with the ground control station. It enables UAVs to fly independently but keeps the operator updated on the mission progress.

Mission Computing

1. Companion Computer

Secondary onboard computer that performs calculations requiring more processing power than those performed by the flight controller. It enables mapping, image processing, obstacle avoidance, sensor data processing, and autonomous mission planning.

2. Data Logger

Stores information gathered during the mission: GPS position, altitude, speed, battery state, flight conditions, sensor data, etc. This information can later be used in analysis.

3. Memory Card (SSD)

Enables storage of large amounts of data gathered during missions: RGB and thermal images, multispectral data, LiDAR scans, flight logs.

4. Real Time Clock (RTC)

Provides onboard system with precise time data. It is important because it makes it possible to synchronize photographs, sensor data, and other information collected during flights with their corresponding time, which is crucial for map creation and analysis.

5. Payload Interface

It serves as an interface between mining sensors and the onboard computer and enables receiving, processing, and storing data coming from cameras, LiDAR, magnetometers, and other payloads.

Communication System

6. Radio Receiver

Receives control commands sent from the operator and ground control station. It enables the operator to control UAV and change some flight commands and modes.

7. Telemetry Radio

Transmits data related to flight parameters from UAV to ground control station: position, altitude, speed, battery state, system status, etc.

8. Video Transmitter

Sends live video feed from onboard cameras to the ground control station. The operator is able to monitor surrounding environment and the mission itself through it.

9. Data Modem

Transfers mission and sensor data between UAV and the ground control station. It allows for transmitting information received from mining payloads.

10. Antennas

Enable wireless communication for control, telemetry, video, and GNSS. Their placement is crucial for successful data transmission.

11Ground Control Connection

This connection links the UAV with the ground control station of the operator. Here, an operator is able to plan missions, monitor the UAV and view flight details and collected data.

10.6. Payload Systems

- ✓ RGB Camera : For use in mine mapping, inspection, and documentation.
- ✓ Thermal Camera : Can detect heat, fires, and equipment malfunctions.
- ✓ Multispectral Camera : Provides assistance in geological and environmental analyses.
- ✓ LiDAR Scanner : Creates 3D maps of mining zones.
- ✓ Magnetometer : Helps to detect magnetic differences for mineral prospecting.
- ✓ Water Quality Sensors : Measures water quality including temperature and pH.
- ✓ Sampling Pump and Tank : Samples water or sediment.
- ✓ Gas Sensors : For detecting harmful gases within the mining zone.

Remote Sensing Compared to Ground Geophysics

The GEOMINER 1 payload performs aerial remote sensing only , it does not perform ground contact geophysical methods such as seismic refraction, electrical resistivity, or borehole logging. The value of the aerial payload is rapid, wide area coverage and high resolution surface mapping, not subsurface volumetric characterisation beyond what spectral, magnetic, or thermal signatures can infer.

10.6.1. Selected Payload Modules

The modular bay accepts packages up to 5kg via a standardised mechanical, power, and data interface. Approved modules for the first-generation system:

Sensor	mass	specification	Specification basis

Survey grade LiDAR	1.2kg	128 beamor solid state, ≥300kpts/s	3D terrain mapping, stockpile volumetrics, canopy penetration
RGB mapping camera	0.8kg	42MP,global shutter, APS-C	Orthomosaic generation, photogrammetric mapping, visual Inspection
Hyperspectral imager	1.5kg	VNIRor SWIR, ≥200 bands	Mineral identification, alteration mapping, lithological discrimination
Multispectral camera	0.4kg	5–10 bands, calibrated radiometry	Vegetation health, landdisturbance monitoring
Thermal camera	0.5kg	Uncooled microbolometer, 640×512	Geothermalanomaly detection, equipment hotspot inspection
Magnetometer	0.6kg	Fluxgate or Alkali vapour, 0.1nT resolution	Magnetic anomaly detection for iron nickel and base-metal exploration
Multi gas sensor	0.3kg	Electrochemical / NDIR, CH_4, CO, H_2S	Hazardous gas plume detection, environmental compliance
Radiation detector	0.4kg	Gamma spectrometer, CsI or NaI crystal	Radioactive mineral zone survey

Table 1.6 : selected payload modules

10.7. Safety and Emergency System

- ✓ Failsafe System : Takes emergency actions when there is a battery or communication failure.
- ✓ Obstacle Sensors : Can detect obstacles.
- ✓ Leak Sensors : Can detect when water leaks into the electronics bay.
- ✓ Temperature Sensors : Monitors the temperatures of vital components.
- ✓ Emergency Thruster Shutdown : Stops water thrusters when there is an obstruction.
- ✓ Water Contact Sensors : Detects water contact on its surface.
- ✓ Protection from Electrical Failures : Provides protection from any kind of electrical fault or overload.

11.Functional Block Diagram Subsystem Description

The system architecture is broken down into five interconnected subsystems, defining how power, data, and commands flow through the UAV:

1. Power & Propulsion Subsystem
   - ✓ Main Battery Pack: High capacity high voltage (HV) LiPo/Li Ion energy storage system supplying total system power.
   - ✓ Power Distribution Board (PDB): Splits main battery power into high-current paths for the motor ESCs and stepped down low voltage (LV) paths for avionics.
   - ✓ VTOL & Cruise ESCs/Motors: Electronic Speed Controllers taking PWM signals from the autopilot to drive the four vertical lift motors and the single rear pusher cruise motor.
2. Avionics & Flight Control Subsystem
   - ✓ Pixhawk Flight Controller: The primary autopilot running firmware (PX4 or ArduPilot) to compute stabilization, autonomous waypoint navigation, and transition logic.
   - ✓ GNSS/RTK Module: High precision satellite navigation system ensuring centimeter-accurate geotagging for geological surveys.
   - ✓ Digital Airspeed Sensor: Measures dynamic pressure via the Pitot tube to give the autopilot reliable airspeed data during transition and cruise.
   - ✓ IMU/Compass Modules: Internal and external inertial measurement units tracking aircraft orientation, acceleration, and magnetic heading.

- ✓ Airborne Telemetry Radio: Wireless RF transceiver linking the aircraft's flight data in real time to the ground station.

3. Actuation & Airframe Subsystem

- ✓ Servo Controller: Dedicated signal distributor routing control outputs from the autopilot. Control Surface Servos: High torque digital actuators driving the ailerons, elevators, and rudders.
- ✓ Retractable Landing Gear Servos: Mechanical actuators used to retract or deploy the landing gear if configured for aerodynamic streamlining during cruise.

4 Payload & Mission Subsystem

- ✓ Modular Payload Bay: Isolated power and mounting bay designed for quick swapping of mission specific hardware.
- ✓ Interchangeable Mining Sensors: Supports LiDAR (for terrain and topographic mapping), Multispectral Cameras (for mineral composition analysis), and Magnetometers (for subsurface geological anomaly detection).
- ✓ Gimbal Camera Unit & Data Logger: Stabilized optical camera system and onboard solid state storage recording raw mission data.

5 Ground Control Station (GCS) Subsystem

- ✓ GCS Laptop/Tablet: Ground terminal running QGroundControl Software for mission planning, monitoring flight parameters, and setting geofences.
- ✓ Ground Telemetry Radio: Ground-side RF module establishing a reliable two-way data link with the aircraft.
- ✓ Q GroundControl Interface / Displays: Real-time software views showing flight telemetry data logs, battery status, and live video feeds from the UAV payload.

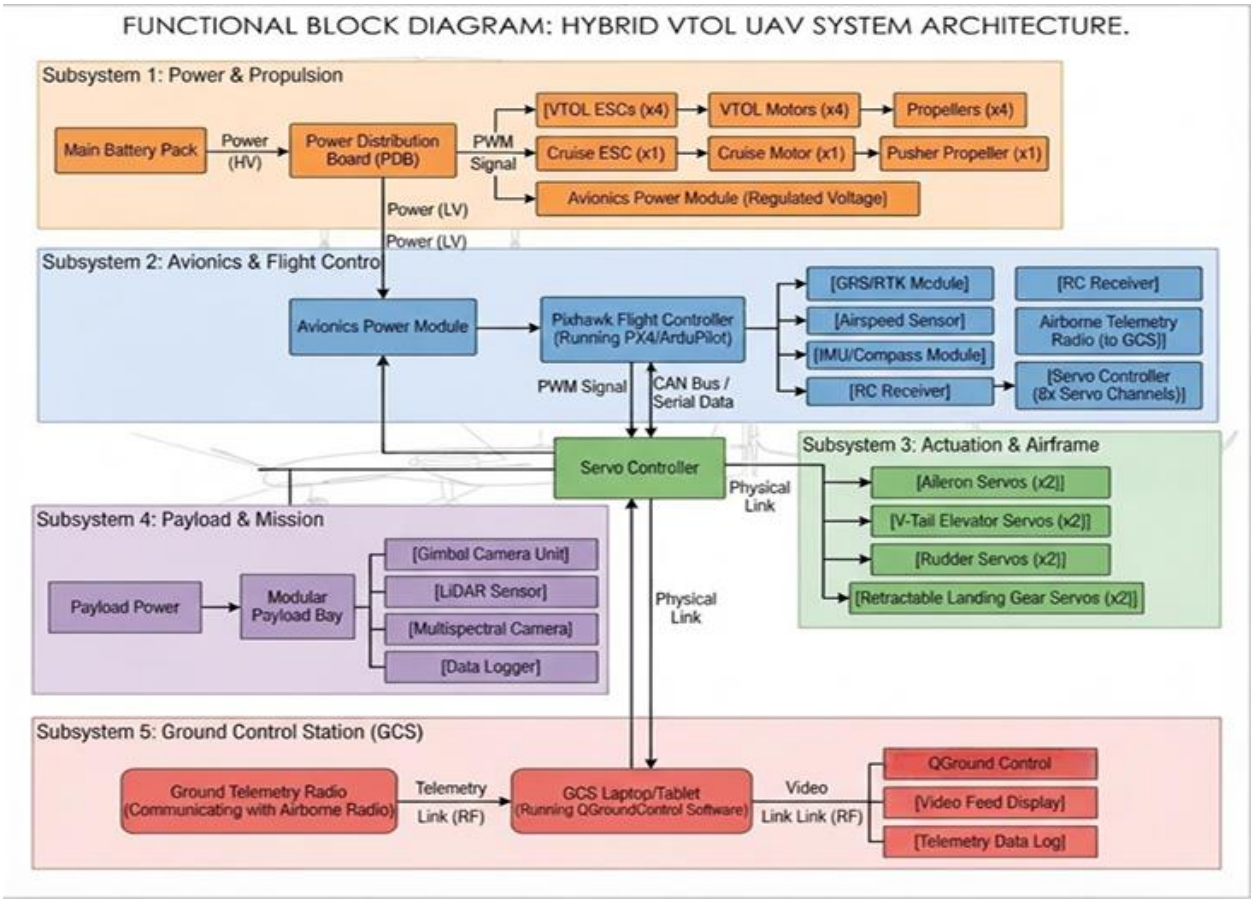


Fig 1.1: functional block diagram

12.Integration Logic

The Pixhawk flight controller serves as the primary real time integration node for the airborne vehicle. It collects attitude and rate information from the IMU, position and velocity corrections from the RTK GNSS module, airspeed from the pitot static sensor, and obstacle proximity information from forward and downward range finders. It processes the ArduPilot QuadPlane control laws and controls the five electronic speed controllers.

The companion computer connects to the flight controller via a dedicated serial or Ethernet link, independent of the flight critical sensor bus. It handles non real time mission logic, payload data intake, optional AI preprocessing onboard, and higher-speed telemetry proxy. Failure of the companion computer will not affect the flight controller.

The ground station QGroundControl communicates with the aircraft using the MAVLink

telemetry radio link. It serves mission planning, real time observation, manual override, and log analysis. The loss of the telemetry link in itself will not end the current autonomous mission: the aircraft will continue along its waypoints and perform failsafe actions.

13.Discussion of Safety , redundancy and mission reliability

13.1. Safety Engineering

Proposed UAV is a VTOL fixed wing aircraft for mineral exploration and mining that should be operated in difficult conditions. Being autonomous and far from the ground control station, it requires special safety systems to decrease the possibility of collision, loss of control, failure of equipment, or damage to people and property.

1.Flight Safety and Failsafe Systems

The flight control system continually controls the status of such systems of the UAV as flight, navigation, power, propulsion, communications, etc. When any unsafe condition is found, corresponding failsafe procedure is initiated automatically.

- ✓ Return to Home (RTH):In case any conditions occur (for example, communication loss, low battery), making the further mission unsafe, the UAV automatically returns home to a previously set location.
- ✓ Geofencing:Software-based geofence defines operating horizontal and vertical limits. Geofence is uploaded before every mission using the ground control system. Breach of the geofence leads to automatic execution of RTH procedure or even to flight termination in case of severe situation.
- ✓ Low Battery Failsafe:State of charge, voltage, current, and the remaining energy of the battery are monitored all the time. When the calculated remaining energy gets the level of return margin, the UAV stops survey and starts returning home. In case of critical situation, it executes corresponding emergency landing procedure.
- ✓ Communication Loss Failsafe:In case of loss of communication with the ground-control station, the UAV automatically switches to a previously defined safe mode (RTH, controlled loitering, or landing).
- ✓ Altitude Limitation:Flight control system keeps the UAV within its operational altitude limits to minimize the risk of entering in any unsafe airspace or terrain.

Pre flight Safety Checks:

Before takeoff, the system and the operator check such UAV systems and components as airframe, motors, propellers, battery, ESCs, flight controller, GNSS, sensors, communications, payload, storage capacity, mission plan, geofence, weather conditions, and emergency landing places.

2.Collision Avoidance

The UAV uses obstacle awareness systems to minimize the risk of collision during lowaltitude flights.

Forward facing LiDAR or stereo vision sensors detect obstacles. Downward facing range sensors provide terrain following and precise landing. Targeted forward detection range is about 50–100 meters, which gives enough time reaction in case of obstacle detection taking into account the design cruise speed of the UAV. If the obstacle is detected, the UAV can slow down, stop, change the route of flight or execute failsafe procedure.

3 Ballistic Recovery Parachute

Ballistic recovery parachute is included as a primary emergency recovery system. In case of a catastrophic failure during the fixed wing survey flight (loss of aircraft control, etc.), it will allow for parachute deployment at the proper altitude and aircraft attitude. The parachute will be armed automatically after switching to fixed-wing cruise mode. It may be triggered by the flight controller itself, Flight Termination System, or operator input. The purpose of the parachute system is to slow down the descent rate and lessen the impact energy on touchdown, decreasing both the UAV and nearby environment damage.

4 Flight Termination System (FTS)

Flight Termination System allows having an additional emergency command channel independent from regular flight controller's failsafe algorithm. The system can be manually triggered by an operator or automatically under specific circumstances (e.g., critical geofencing breach).

5. Controlled Emergency Descent

In case of catastrophic failure of the UAV below the minimum parachute deployment altitude, the UAV shouldn't deploy the parachute where there wouldn't be enough time left

for its full inflation and stabilization. In such a situation, the UAV should utilize any existing VTOL/flight control capabilities for the most controlled possible descent.

6 Emergency Water Landing

As the UAV could be used near water bodies during mining and mineral exploration missions, emergency water landing system is included in the UAV safety concept. The UAV is able to utilize the floats, buoyancy of fuselage and/or any other flotation devices in order to have positive buoyancy after landing in water. Sensitive electronics and battery compartments should be protected from the water intrusion. Flight control systems should maintain a stable attitude on the descent and touchdown if possible.

7 Environmental Safety

The UAV is capable of operating under specified environmental constraints such as:

- ✓ Wind: Ability to function within specified wind limit.
- ✓ Dust and moisture: Protection of electronic systems and connections from dust and moisture.
- ✓ Temperature: Functioning within specified temperature limits.
- ✓ Vibration: Insulation and protection of electronics and sensors against vibration.
- ✓ Uneven surfaces: Selecting appropriate launch and landing locations.
- ✓ The UAV should not operate if environmental conditions exceed its designed limitations.

Emergency Response Hierarchy

- ✓ Emergency response involves hierarchy as follows:
- ✓ Operator override → Manual hold, RTH or FTS.
- ✓ Geofence violation → RTH or flight termination depending on severity.
- ✓ Low battery → Mission termination and RTH.
- ✓ Critical failure → Deployment of parachute if conditions permit.
- ✓ Critical failure at low altitude → Descent using available control authority.
- ✓ Ground landing site unsafe → Water landing if there is a nearby water surface and UAV can float.

13.2. Redundancy and Reliability

Redundancy is the application of multiple or alternative systems in such a way that the failure of any of them will not immediately cause the loss of the entire UAV. For the hybrid VTOL

UAV intended for long endurance operations in remote mining areas, this aspect is especially critical.

- ✓ Sensor Redundancy

The UAV uses several navigation and flight state sensors, namely:IMU,GNSS,Barometer,Magnetometer,Airspeedsensor,RangefinderorLiDAR.
Flight controller verifies sensor information. In case of abnormal sensor measurement, information from other sensors will be available and will allow the UAV to safely operate.

- ✓ Propulsion Redundancy

The hybrid VTOL design implies the use of multiple vertical lift propulsion systems and an independent cruise propulsion system. The propulsion system configuration provides some fault tolerance.
Monitoring of each propulsion unit includes:Motor performance,ESC condition,Current consumption,Battery voltage,Temperature and Propulsion imbalance
If there is enough control authority left, the UAV must be able to maintain controlled flight after a limited number of propulsion failures.

- ✓ Power Redundancy

Critical systems of the UAV include the flight controller, navigation sensors, communication system, and emergency systems. Thus, they need a reliable power source.
The power management system monitors:Battery voltage , Current ,Over current condition, Short circuit condition,Battery condition,Temperature and Remaining energy
Emergency energy reserve must be provided to have sufficient energy for RTH, controlled descent, and emergency landing.

- ✓ Communication Redundancy

The main communication channel allows the UAV to communicate with the ground control station. If mission requires it, alternative communication channels like cellular or satellite communication are applied.
The system monitors signal strength and communication quality. Loss of the main communication channel results in activation of predefined communication failsafe.

- ✓ Data Redundancy

Data on geology and mineral survey are critical mission products. Therefore, data needs protection from the loss of communication and storage failure. Protection is provided by: Onboard data storage,Automatic data recording,Sufficient data storage capacity,Backup of important mission data and Data transfer from onboard storage to the ground station if available

Data recording onboard the UAV is performed even if there is no connection with the ground station.

13.3. Mission Reliability

Mission reliability refers to the capability of the UAV to safely accomplish its mission without errors and produce required quality of geological data.

Before the mission, all of the airframe, propulsion, battery, flight control system, GNSS, sensors, communications, payload, mission plan, geofencing, weather, storage capacity and emergency landing sites of the UAV are checked by the operator.

During the mission, the UAV keeps tracking its position, altitude, speed, attitude, battery status, GNSS integrity, communication status, propulsion status, sensors status and payload status.

This system works in general flow of:

Normal state => Warning => Critical state =>Failsafe => Safe return

After the mission, the airframe, propellers, motor, battery, payload, sensors, flight logs and geological data obtained by the UAV are checked. Flights logs are analyzed to detect any faults or abnormal states which may influence future missions.

Overall, the use of failsafe systems, parachute system, flight termination system, controlled descent, emergency landing in water, collision avoidance, environmental protection and system redundancy increases reliability and safety of hybrid VTOL UAV in remote mining and mineral exploration missions.

14. Material Selection And Engineering Justification

1. Material Selection Objective

The material selection for the 30 kg hybrid VTOL UAV focuses on achieving a lightweight, strong, stiff, durable, non-corrosive, and maintainable airframe.

The selected materials must withstand aerodynamic loads, VTOL landing loads, repeated loading, environmental exposure, and field operation in rugged mining environments while keeping structural mass as low as possible.

The main component selection criteria are:

- ✓ Low density
- ✓ High strength
- ✓ High stiffness
- ✓ Impact resistance
- ✓ Corrosion and environmental resistance
- ✓ Cost effective and availability

Primary Material Selection

Carbon Fiber for the airframe , Carbon fiber is selected as the primary structural material for the UAV.

Proposed application : Main wing ,Wing spars ,Wing skins , Fuselage , Structural plates, VTOL support structures and Major load-bearing components

Engineering justification

Carbon fiber has a high strength and high stiffness. This is particularly important for a 30 kg long-endurance aircraft because unnecessary structural mass increases the required lift and propulsion energy. It is strong, light weight, corrosion resistant, and also it has high stiffness.

It’s high stiffness also helps maintain the designed aerodynamic shape of the wing and reduces structural deformation during flight.

Carbon fiber also provides good resistance to fatigue and does not corrode like conventional metals, making it suitable for repeated missions and outdoor operation.

Secondary Material :Aluminum alloy is selected as a secondary structural material, not as the main airframe material.

Proposed applications : Motor mounting interface , Structural brackets , Hinges ,Equipment mounting plates , Local reinforcementand Mechanical joints

Engineering justification

Aluminum has good strength, excellent machinability, and predictable mechanical properties. It is suitable for areas where concentrated loads or mechanical joints make composite construction less convenient.

However, using aluminum for the complete airframe would increase structural mass compared with a carbon fiber based structure.

The material selection is driven primarily by the UAV's 30 kg MTOW and long-endurance mission.

Carbon fiber is the preferred primary material because reducing structural mass directly improves the aircraft's payload capability and energy efficiency. Its high stiffness also helps maintain wing and fuselage structural integrity during cruise and VTOL operations.

15.mass budget

This is a bottom-up mass budget. Every line item is either a stated component-class mass estimate consistent with the commercial component class selected elsewhere, or a calculated derived value. The battery size is set to meet the 120-minute practical endurance requirement, and the resulting battery mass is included as a component like any other. The sum closes at 30kg MTOW with an explicit structural margin.

Item	Mass(G)	Notes
Fuselage (CFRP shell, formers)	2800	Laminatedcarbonfibrewith printed internal ribs
Main wing (spars, ribs, skin)	1800	Two spar CFRP construction
Tail booms (2)	600	Pultruded CFRP tube
Empennage	450	CFRP skin and ribs
Landing gear (4 legs)	800	Aluminium strut, elastomer foot, bolton
VTOL arms (4)	1200	CFRP or CF-nylon with al minium root brackets
VTOL motors (4)	1400	≈350g each, large frame class
VTOL propellers (4)	800	22in folding or fixed
Pusher cruise motor	400	Fixed wing outrunner/inrunner
Cruise propeller (folding)	300	18in two blade folding
ESCs (5 total)	700	4×60A VTOL + 1×80A cruise
Main battery (14S Li ion)	10000	Sized for 120min mission energy
Backup battery	200	Small Li ion for avionics survival
Flight controller , IMU	150	Pixhawk standard with dual IMU
RTK GNSS rover , antenna	200	Multi-band receiver
Magnetometer,barometer, Airspeed	100	Combined sensor module
Companion computer	200	Jetson class module and carrier
Telemetry radio, antennas	300	Longrange 20km class
Obstacle sensors	250	Forward and downward rangefinders

Power distribution ,wiring	800	PDB, DC to DC, harness, connectors
Payload bay structure	400	Rails, locking mechanism, enclosure
Three-axis gimbal	600	Brushless stabilized
Parachute, FTS , beacon	500	Recovery system, termination, locator
Fasteners, adhesive, misc	350	M3/M4 hardware, bonding, tape
Subtotal(airframe, avionics , propulsion)	25250	
Payload (max)	5000	Modular sensor package
Subtotal with max payload	30250	
Design margin (adjustment)	−250	Trim to exact MTOW via battery or payload management
MTOW	30000(30 kg)	

Table 1.7: mass budget

The battery is approximately 33percent of MTOW a direct, disclosed consequence of the 120 minute endurance and 100km range requirements. A smaller battery would reduce mass but fail to meet the energy requirement; this trade is accepted because endurance and range are specified as firm requirements.

16. Power Budget and Endurance

16.1.Mission Segment Power

Load	Hover(W)	Cruise(W)	Notes
VTOL propulsion (4 motors)	4336	0	Momentum-theory estimate, FM = 0.75
Cruise propulsion (pusher)	0	1806	XFLR5 drag estimate, design baseline
Flight controller + IMU	5	5	
Companion computer	15	15	Active SLAM/AI load
RTK GNSS + magnetometer	8	8	
Airspeed + barometer	2	2	
Telemetry radio	10	10	
Obstacle sensors	8	8	
Gimbal (active stabilisation)	25	25	
Payload (typical survey)	85	85	LiDAR + RGB + thermal typical mix
DC-DC conversion loss (est.)	50	50	5% of total load estimate
Total segment power	4534	2014	

Table 1.8: Mission Segment Power

16.2 Endurance Calculation

The mission profile is defined as 6min VTOL (take-off, transition, landing combined) plus 114min cruise, totalling 120min. Energy per segment:

$$E_{VTOL} = 4.534\text{kW} \times 0.10\text{h} = 0.453\text{kWh} \quad (9)$$

$$E_{cruise} = 2.014\text{kW} \times 1.90\text{h} = 3.827\text{kWh} \quad (10)$$

$$E\text{total, used} = 4.280\text{kWh}. \quad (11)$$

With a 20% reserve (standard practice for Li-ion mission planning, preserving cell longevity and providing contingency):

$$E_{\text{battery, min}} = \frac{4.280}{0.80} = 5.35\text{kWh}. \quad (12)$$

At 14S (51.8V nominal), the required capacity is:

$$C_{min} = \frac{5.35\text{kWh}}{51.8\text{V}} = 103.3\text{Ah}. \quad (13)$$

The selected battery is a 14S 110Ah Li-ion pack, providing:

$$E_{installed} = 51.8\text{V} \times 110\text{Ah} = 5.70\text{kWh}. \quad (14)$$

This closes the practical mission endurance requirement with margin. Actual usable endurance depends on real drag, propulsive efficiency, wind, and temperature, and is confirmed in the endurance test.

16.3. Range Verification

At the design cruise speed of 20m/s (72km/h), the cruise-only range on the usable cruise energy (after VTOL segments and reserve) is:

$$R_{cruise} = 20\text{m/s} \times 1.90\text{h} \times 3600\text{s/h} = 136.8\text{km}. \quad (15)$$

After accounting for headwind, transition manoeuvres, and loiter margin, the 100km design mission range is achievable within the installed energy, confirmed by the same endurance test.

17.Design and engineering software and simulation reqirements

3D CAD Modeling

A 3D model of the UAV airframe was created using Onshape. The model features the major structural parts of the aircraft, including the fuselage, wings, motor mounts, landing gear, and mounting hardware for carrying cargo on board. Additionally, OpenVSP was used to assist with the conceptual aerodynamic geometry and overall configuration of the aircraft, whereas XFLR5 was used for preliminary aerodynamic analysis of the fixed wing configuration.

Ground Station and Flight Simulation

QGroundControl is the chosen software package for UAV control from the ground station that allows for the planning of missions and monitoring of various parameters of the aircraft's operation. This software may be used in combination with the Pixhawk flight controller to plan missions and track the major aircraft's parameters. Nonetheless, since there was no physical autopilot at hand, QGroundControl was mainly used for simulations and mission planning.

Simulation Requirements for This UAV For the 30 kg hybrid VTOL, simulation should cover both engineering behavior and autonomous flight behavior.

- ✓ Aerodynamic simulation Estimate lift, drag, pressure distribution and cruise performance; investigate wing and fuselage geometrty.
- ✓ Structural simulation Apply representative wing, fuselage, landing and motor-mount loads; identify high-stress areas and deformation.
- ✓ Propulsion simulation Check motor/propeller thrust, power, current and efficiency against hover and cruise requirements.
- ✓ Mass and CG study Evaluate empty mass, battery/fuel mass, payload variations and center-of-gravity limits.
- ✓ Energy/endurance model Estimate energy consumption in VTOL, transition and cruise phases and compare with the 2-hour target.

- ✓ Flight-control SITL Test modes, sensor behavior, actuator commands, takeoff, transition, cruise, return and landing without risking the physical UAV.
- ✓ Mission simulation Build a representative mining/geological survey route in QGroundControl and test autonomous waypoint behavior.
- ✓ Failure/failsafe tests Test loss of telemetry, low battery, GNSS issues, sensor faults and other defined safety responses in simulation where supported.

18.Engineering Trade-Off Analysis

✓ Hybrid VTOL versus Pure Multirotor or Pure Fixed-Wing

A pure multirotor at 30kg MTOW requires approximately 8–10kW continuous hover power to achieve 100km range, demanding an impractical battery mass (>20kg) and yielding flight times well under 60minutes. A pure fixed-wing meets endurance and range but requires a runway or catapult, eliminating operation from unprepared remote sites. The hybrid VTOL configuration accepts the mass and complexity penalty of four VTOL arms and motors in exchange for runway independence plus cruise efficiency.

✓ Carbon Fibre versus Aluminium Primary Structure

Carbon fibre composite is selected for the primary structure (wing spars, fuselage, booms): its specific stiffness and strength exceed aluminium at the 2.8m span and 30kg scale, and bonded composite joints eliminate the stress concentrations inherent to riveted or bolted aluminium sheet. Aluminium is retained for high-wear items (landing gear struts, motor-mount backing plates) and components requiring machined precision.

✓ Modular versus Fixed Payload

A fixed payload would simplify structure and wiring but restrict the aircraft to a single sensor type per campaign. The modular quick-release bay adds 0.4kg of rails and connector mass but enables the same airframe to perform LiDAR mapping, hyperspectral

mineral identification, and gas-detection missions on consecutive sorties without airframe modification – flexibility judged essential for a multi-mission mining platform.

- ✓ 14S Li-ion versus 12S LiPo

12S lithium-polymer (LiPo) is common in heavy-lift multirotors but carries lower gravimetric energy density (typically 150–180Wh/kg) than modern Li-ion pouch cells (220–260Wh/kg). For the 5.7kWh energy requirement, LiPo would add approximately 3-4kg of battery mass, breaking the 30kg MTOW closure. 14S Li-ion is selected despite higher per cell complexity and BMS cost.

- ✓ Ballistic Parachute versus Controlled Descent Only

At 30kg and typical survey altitudes above 50m, a ballistic parachute delivers a meaningful reduction in impact energy and ground hazard compared to an uncontrolled crash. The mass penalty (approximately 0.4kg for the parachute and 0.1kg for the deployment mechanism) is accepted as a non-optional safety feature for over-land operation in remote terrain where emergency-services response is delayed.

19.Performance specifications and expected operational capabilities

Parameter	Value	traceability
Configuration	Hybrid VTOL QuadPlane	Requirement
MTOW	30kg	Requirement, bottom-up budget
Payload capacity	up to 5kg	Design target
Endurance	≥120min practical	Requirement, energy budget

Range	100km	Design target

Cruise speed	18–25m/s	Design target
Max speed	30m/s	Design target
Wingspan	2.8m	Design target
Length	1.9m	Derived
Hover T/W	≥1.8	Requirement
VTOL propeller	22in	Selected component class
Cruise propeller	18in folding	Selected component class
Battery	14S 110Ah Li-ion	Derived from energy budget
Main battery mass	10.0kg	Component estimate
Navigation	RTK GNSS + EKF3	Requirement
RTK accuracy	<5cm horizontal	Requirement
Autopilot	ArduPilot QuadPlane on Pixhawk	Requirement
GCS	QGroundControl	Requirement
Recovery	Ballistic parachute + FTS	Requirement
Operating env.	Daylight VMC, −5 to +40°C, dry	Requirement / Assumption
Max wind	12m/s sustained	Design target / TBD
Service ceiling	4000m ASL	Assumption

Table 1.9: Performance specifications

Expexted Operational Capability

The UAV is expected to provide the following operational capabilities:

- ✓ VTOL (Vertical Take-Off and Landing): The UAV is capable of taking off and landing vertically without need for conventional runway; therefore, it can work in remote mining areas where there is no suitable runway.
- ✓ Fixed Wing Cruise: After transition from VTOL to fixed wing operation mode, the UAV can fly longer distances with extended endurance capability.
- ✓ Mineral Exploration: The UAV is capable of carrying and working with payload equipment including RGB camera, multispectral, hyperspectral, LiDAR, thermal and magnetic sensors for geological survey and mineral exploration.

- ✓ Accurate Mapping: The UAV can be used for accurate mapping due to its RTK/PPK GNSS and imaging sensors.
- ✓ Terrain Monitoring: The UAV can use aerial photos in order to monitor the area of excavations, stockpiles, slopes, roadways and other mining facilities.
- ✓ Rugged Environment Operations: The UAV is designed to operate in rugged environments with limited facilities, some wind and dust presence, temperature changes and elevated terrain.
- ✓ Autonomous Operation: The UAV will follow pre planned flight path, gather data from payload sensors, monitor flight parameters and return back to home when needed. Safety and Emergency Operation: The UAV has built in safety functions including RTH (Return To Home), battery monitoring, communication loss function, obstacle detection and emergency landing.
- ✓ Aerial Water Adjacent Operations: Using special buoyant structures, the UAV can perform emergency water landing and be used for aerial surveys near lakes, rivers or offshore mineral exploration.
- ✓ Flexible Payload Bay Operation: The UAV can operate different payload modules depending on requirements of exploration, surveying, mapping and environmental monitoring missions.

20.Future improvements

- ✓ Contact Geophysics : Towed/dropped magnetotelluric/seismic sensor module for subsurface investigations.
- ✓ Onboard AI Mineral Classification : Onboard edge AI will provide real time mineral classification and anomaly detection decreasing the processing time.
- ✓ Swarming capabilities of multiple UAVs can help in mapping of large mines at once using one RTK base station.
- ✓ Satellite Communications : L band satellite modem can ensure beyond line of sight communications in remote locations.

- ✓ Weather Resistance : Enhanced sealing of electronic components and payload enclosure will make the system resistant to dust and rain providing IP54 class of protection.
- ✓ Dual Flight Controller : System with dual flight controller having automatic failover can increase reliability of flights preventing controller crashes.
- ✓ Higher Energy Density : Future solid state or lithium sulfur batteries can increase energy storage and prolong flight endurance of the system.
- ✓ Solar Energy : Specially designed lightweight solar panels can be installed on the wings providing extra electricity and flight endurance in appropriate lighting conditions.
- ✓ Water Applications : UAV can be modified to conduct water missions adding appropriate floatation, water resistant electronics and water monitoring payload.

21.Bill of Materials

Preliminary bill of materials. Where a specific commercial part number would be premature to fix, a performance class and representative alternative are given.

Part	Qty	specification	Function
VTOL motor	4	Largeframe multirotor, ≈350g, 22in rated, 14S	Hover and VTOL Thrust
VTOL propeller	4	22in, ≈8in pitch, two-blade, CF-reinforced nylon	VTOL thrust generation
Cruise motor	1	Fixed wing out runner/inrunner, ≈400g, 18in rated, 14S	Cruise propulsion

Part	Qty	specification	Function
Cruise propeller	1	18in,two blade, Folding	Cruise thrust generation
ESC (VTOL)	4	60A continuous, 14S rated, with BEC	VTOL motor control
ESC (cruise)	1	80A continuous, 14S rated	Cruise motor control
Flight controller	1	Pixhawk-standard (e.g., Cube Orange+)	Autopilot hardware
RTK GNSS Rover	1	Multi-band, RTCM3 input, with antenna	Centimetre positioning
RTK GNSS base	1	Multi-band, surveygrade with tripod	Base-station corrections
Companion computer	1	Jetson Orin Nano or equivalent	Onboard compute
Telemetry radio	1	20km range, MAVLink compatible	GCS data link
Main battery	1	14S 110Ah Li ion, 5.7kWh, with BMS	Primary energy storage
Backup battery	1	4S 2Ah Li ion	Avionicssurvival power
PDB	1	14S rated,200A, with current sensing	Power distribution
DC-DC con-Verter	1	Isolated,14S to 5V/12V, 200W	Avionics/payload bus
Airspeed sensor	1	Digital pitot-static, I2C/UART	True airspeedmeasurement
Magnetometer	1	External, I2C, away from power wiring	Heading reference
Barometer	1	Integrated or external, I2C/SPI	Altitude reference
Forward obstacle sensor	1	Lidar or radar, 100m range	Collision avoidance
Downward Rangefinder	1	Lidar, 50m range	Terrainfollowing or Landing

Part	Qty	specification	Function
Parachute system	1	Ballistic deployment, 30–35kg rated	Emergency recovery
FTS	1	Independent motor cutoff circuit	Flight termination
Locator beacon	1	406MHz or AIS, battery-backed	Post incident recovery
Gimbal	1	Three-axis brushless, 1kg payload capacity	Sensor stabilization
Payload modules	As req.	LiDAR,RGB, hyperspectral, thermal, gas, magnetometer	Mission sensing
CFRP tube/plate	As req.	Various diameters and thicknesses	Primary structure
PETG/nylon filament	As req.	1.75mm, engineering grade	Printed formers and Covers
Fasteners	As req.	M3/M4 stainless, heat-set brass inserts	Assembly joints

Table 1.10: Preliminary Bill of Materials

conclusion

GEOMINER-1 is the conceptual design of an unmanned aerial vehicle weighing 30 kg with VTOL capabilities intended to explore minerals, survey geology, map mines, and monitor the environment in remote territories.

The combination of modular payload, a flight control system with Pixhawk, RTK-GNSS guidance, and several other safety solutions provides a good basis for further developments. At present, the stated performance values are the design objectives; however, further research is needed, which includes aerodynamic analysis, simulation, prototyping, and actual flight tests. Overall, GEOMINER-1 can be considered as a good choice for safe and efficient surveying operations in difficult mining conditions.

Reference

- ✓ Background Aerodynamics knowledge
- ✓ https://share.google/wXlCyn4LICdPf8Ggu
- ✓ https://share.google/ht2h8QmqHmclvDIsr